



Xi'an Jiaotong-Liverpool University

西交利物浦大學

FIN302

Finance and Markets

Lecture 7: Derivative Markets

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An Overview

➤ Contents

- Derivative markets and futures hedging (Lect 7-8)
- Swaps (Lect 9)
- Options, pricing, and hedging (Lect 10-12)

➤ Economic Intuitions are your starting point!

- Trade-off between benefits and costs; a better return for high-risk investment
- You pay what you get in the future

➤ Economic mechanisms are crucial for a good understanding

- Pricing: no-arbitrage principle or no free lunch
 - If your investment has no risk, your return should be given by a risk-free rate
 - If you invest nothing, you will get nothing; e.g., your return should be zero



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Good Practice of Learning

➤ Be Ready to Change Your Gear!

- This is an advanced course on derivatives; so prepare yourself!
- Read the slides before the lectures and focus on the part you may have difficulty
- Review lecture slides and focus on the concepts and examples after lectures
- Try tutorial questions before checking the answers
- Come to the **consultation hours, Thursday 3 pm-5 pm (BS583)**, if you need help before it is too late

➤ DO NOT being over-confident

Lecture 7- Derivative Markets

Questions and Learning Outcomes

- What are derivatives?
 - Derivative Markets
 - Examples of Derivatives: Forwards, Futures, Options
- How to use derivatives for hedging and risk management?
 - Hedging Strategies Using Futures
 - Forward / Futures Price:
 - Stock, index, and currency futures

Lecture 7- Derivative Markets

- Readings:
 - Hull (2022), Chapters 1, 2, 5
- Tutorial and Homework Problems from Hull:
 - 1.12, 13, 30
 - 2.11, 17, 21
 - 5.13, 17, 18

Derivatives

A **derivative** is a financial instrument

- Involves **two parties** agreeing to **future transactions**
- whose value depends on the value(s) of other **underlying variables**.

➤ Examples:

- Forward/Futures
- Options
- Swaps
- Interest rate futures/options
- Credit...electricity...weather...insurance
- ...

Derivative Markets

Total Global Assets: US\$ 1,028 Trillions



Using Derivatives

How are derivatives used?

- To hedge risks
- To speculate (take a view on the future direction of the market)
- To lock in an arbitrage profit
- To change the nature of liabilities or investments without incurring the costs of selling one portfolio and buying another portfolio
-

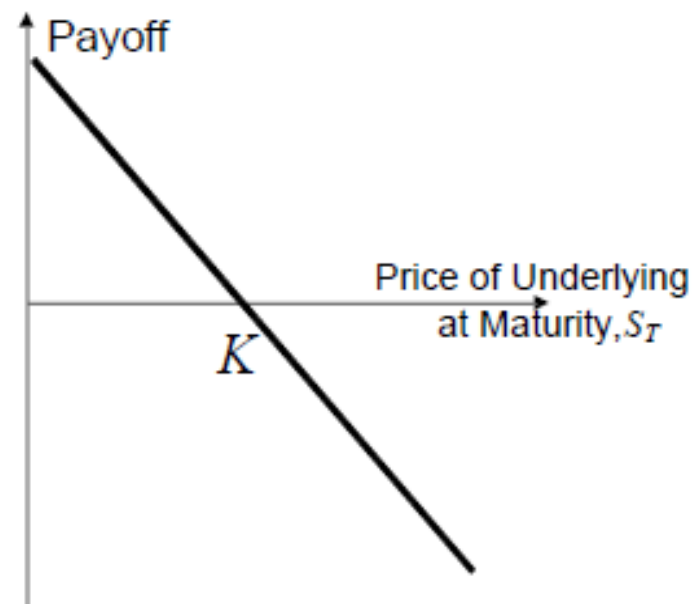
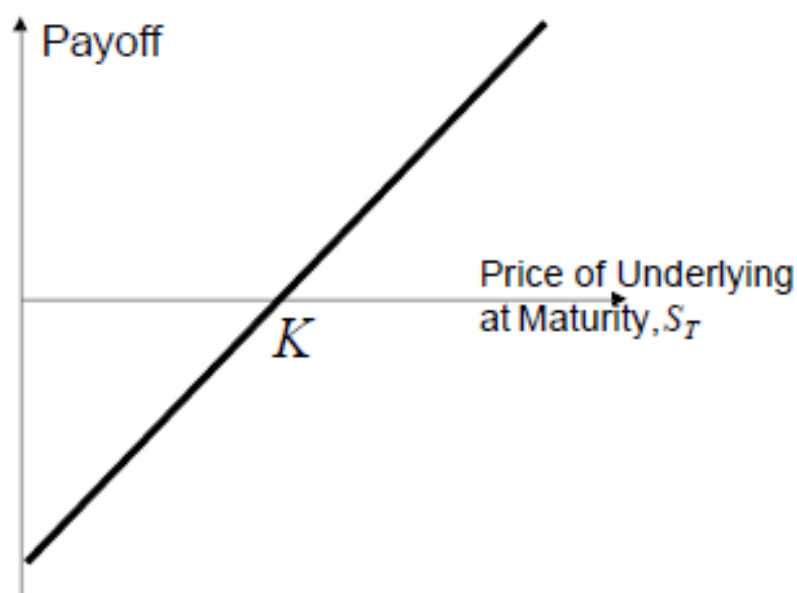
Forward

- A **forward** contract is an agreement between two parties to buy or sell an asset at a certain time (the maturity date T) for a certain price (the forward price or the delivery price K):
 - The party that undertakes to buy has a long position. The payoff of a long forward is $S_T - K$.
 - The party that undertakes to sell has a short position. The payoff of a short forward is $K - S_T$.

Forward

Figure: Payoffs from forward contracts with forward price K and asset price at maturity S_T :

(a) long position: $S_T - K$ (b) short position: $K - S_T$



Hedging with Forwards

Example: On 24 May a US company knows it has to pay £10m on 24 August.

- The company's risk is that the GBP will appreciate. It can hedge this risk by entering a long position in a 3-month forward on £10m at 1.4415 USD/GBP.
- In three months the company will pay \$14,415,000 for its GBP.
 - If the spot exchange rate is 1.3000 on 24 August 2010, then the company loses \$1,415,000 by hedging.
 - If the spot exchange rate is 1.5000 on 24 August 2010, then the company makes a profit of \$585,000 by hedging.
- **Note:** Forward hedging reduces risk by locking in the price. This does not guarantee a better outcome.

Bid/Ask and Spread

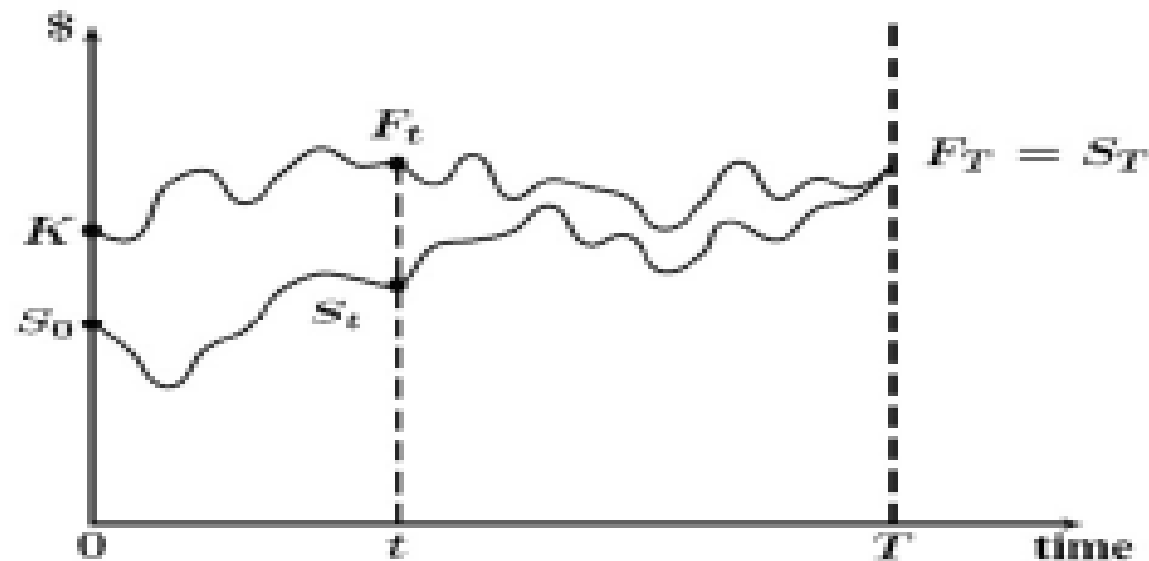
- Quotes are given in pairs that reflect the bid-ask price.
- Example: £1 pound sterling is quoted at \$1.2219-28 by a bank.
 - \$1.2219 is the bid rate at which
 - bank buys pounds from you
 - you sell pounds to the bank
 - \$1.2228 is the ask rate at which
 - bank sells pounds to you
 - you buy pounds from the bank
 - The bid/ask is often quoted by the last two numbers; e.g., 19-28. The bid-ask spread \$0.0009 represents the dealer's profit, which is 9 basis points.
- The spread depends on exchange rate volatility.

Futures

- **Futures** are similar to forwards but are exchange-traded.
- The exchange specifies various **standardized features** of the contract:
 - The underlying asset:
 - The exchange specifies the grade of the underlying commodity for commodities futures.
 - The contract size:
 - If the contract size is too large, then many investors who wish to hedge small exposures cannot use the exchange.
 - If the contract size is too small then trading will be too expensive.

Futures

As the delivery period of a futures contract approaches, the *futures price converges to the spot price* of the underlying asset.



Futures: Trading Risk and Margins

- Margins protect traders from counter-party defaults due to futures price variations.
- An investor maintains a margin account with his broker:
 - The **initial margin** is deposited into this account when the investor first takes a futures position.
 - After each trading day the investor's margin account is **marked to market**.
 - The balance of the investor's margin account is required to always remain above the **maintenance margin**.
 - The investor receives a **margin call** if the balance of his margin account drops below the maintenance margin.
 - He must then deposit the variation margin in his margin account, which returns its balance to the initial margin.
 - The broker closes the investor's position out if he does not post the variation margin.

Futures: Trading Risk and Margins

Example: An investor contacts his broker on 5 June to enter a long position in **two December gold futures contracts**. Each contract is for **100 ounces of gold**, and the futures price on the day is **\$1,250 per ounce**.

- The **initial margin** required by the broker is **\$6,000 per contract**.
- The **maintenance margin** required by the broker is **\$4,500 per contract**.

Day	Trade Price (\$)	Settle Price (\$)	Daily Gain (\$)	Cumul Gain (\$)	Margin Balance (\$)	Margin Call (\$)
1	1,250.00				12,000	
1		1,241.00	-1,800	- 1,800	10,200	
2		1,238.30	-540	- 2,340	9,660	
.....						
6		1,236.20	- 780	-2,760	9,240	
7		1,229.90	- 1,260	- 4,020	7,980	4,020
8		1,230.80	180	-3,840	12,180	
.....						
16	1,226.90		780	-4,620	15,180	

Forward/Futures Price

➤ Asset Classes:

▪ *Investment assets:*

- stocks, bonds, gold, silver (even though it does have some industrial uses)

▪ *Consumption assets:*

- copper, oil, pork bellies

➤ Pricing Mechanism: **No Arbitrage Argument**

- can be used to determine forward and futures prices of *investment assets* from their spot prices,
- but *not for consumption assets*.

➤ Implementation: a combination of long and short positions in futures and the underlying assets

Forward Price: Notations

➤ Notation:

S_0 the spot price of the underlying asset at $t = 0$

S_t the price of the underlying asset at time $t > 0$

F_0 the forward or futures price at $t = 0$;
delivery price of underlying asset negotiated at $t = 0$
for delivery at time T

F_t the forward or futures price at $t > 0$
delivery price of underlying asset negotiated at t for
delivery at time T

r the annual continuously compounded risk-free
interest rate

Forward Price

- A forward contract on a **non-income-paying investment asset**

- Non-dividend-paying stocks, zero-coupon bonds
- The **forward price** of a non-dividend-paying asset

$$F_0 = S_0 e^{rT}, \quad F_t = S_t e^{r(T-t)}$$

- Forward contract on an **investment asset that pays a predictable income stream**, e.g., stocks paying known dividends, coupon-bearing bonds

- The **no-arbitrage forward price**:

$$F_0 = (S_0 - I_0) e^{rT}, \quad F_t = (S_t - I_t) e^{r(T-t)}$$

- I is the present value of all income paid by the asset during the life of the forward.

Forward Price: Known Income

Example: Consider a **10-month forward** on a stock that pays a **dividend of \$0.75 after 3, 6, and 9 months**. Suppose the stock is currently trading for \$50 and the risk-free interest rate is 8% per annum, with continuous compounding, for all maturities.

- The present value of the dividends paid during the life of the contract is

$$I_0 = 0.75e^{-0.08\frac{3}{12}} + 0.75e^{-0.08\frac{6}{12}} + 0.75e^{-0.08\frac{9}{12}} = 2.162$$

- The forward price is therefore

$$F_0 = (S_0 - I_0)e^{0.08\frac{10}{12}} = 51.14$$

- In other words, when the stock is delivered after 10 months the party with the long position will pay \$51.14 per share.

Forward Price: Known Yield

- Forward contract on an investment asset that provides a known yield q , is the average annual continuously compounded
- The no-arbitrage price of a forward contract:

$$F_0 = S_0 e^{(r-q)T}, \quad F_t = S_t e^{(r-q)(T-t)}$$

Example: Consider a 3-month futures contract on an index. Suppose that the stocks underlying the index provide a dividend yield of 1% per annum (cont. comp.), that the current index level is 1,300 and risk-free rate is 5%.

- In this case, $S_0 = 1,300, r = 0.05, q = 0.01, T = 0.25$
- Then $F_0 = S_0 e^{(r-q)T} = 1,300 e^{(0.05-0.01) \times 0.25} = \$1,313.07$

Forward Price: Valuing the Forward

- The **value of a forward contract** and **the forward price** are different.
- When a forward contract is first negotiated, its value is zero, $f_0 = 0$
- However, as time passes, and the initial delivery price of the contract no longer equals the current forward price.
- Consider a trader with a long position in the existing contract and a short position in the new contract at time t :
 - His payoff at time T is therefore
$$(S_T - F_0) + (F_t - S_T) = F_t - F_0$$
 - The present value of his position is thus $(F_t - F_0)e^{-r(T-t)}$
 - The value of his long position at time t :

$$f_t = (F_t - F_0)e^{-r(T-t)}$$

Forward Price: Valuing the Forward

Example: Some time ago a trader entered in a long forward on a non-dividend-paying stock, with a delivery price of \$24. The contract is due to mature after a further 6 months. Suppose the current stock price is \$25 and the risk-free interest rate is 10% per annum, with continuous compounding.

- The current forward price is

$$F_t = S_t e^{r(T-t)} = 25 e^{0.1 \frac{6}{12}} = 26.28$$

- The value of the trader's position is

$$f_t = (F_t - F_0) e^{-r(T-t)} = (26.28 - 24) e^{-0.1 \frac{6}{12}} = 2.17$$

- The value of the position of the trader's counterparty
– 2.17

Futures and Value

- The only difference between futures and forwards is the margining mechanism imposed by futures exchanges.
- When interest rates are deterministic, the no-arbitrage price of a forward contract equals the no-arbitrage price of the equivalent futures contract.
- However, when interest rates are stochastic (i.e. they change randomly over time), forward prices and futures prices do differ—but the difference is generally very small.
- We treat futures contracts as forward contracts, as far as pricing is concerned.
- Due to the *mark-to-market*, **the value of an equivalent long position in futures is**

$$f_t = F_t - F_0$$

Index Futures

- The **futures price of a stock index** is given by

$$F_0 = S_0 e^{(r-q)T}, \quad F_t = S_t e^{(r-q)(T-t)}$$

- q the average annual continuously compounded dividend yield of the portfolio of stocks underlying the index.

Example: Consider a 3-month futures contract on an index. Suppose that the stocks underlying the index provide a dividend yield of 1% per annum (cont. comp.), that the current index level is 1,300 and risk-free rate is 5%.

- In this case, $S_0 = 1,300, r = 0.05, q = 0.01, T = 0.25$
- Then $F_0 = S_0 e^{(r-q)T} = 1,300 e^{(0.05-0.01) \times 0.25} = 1,313.07$



Currency Futures

- For currency futures we take
 - r the domestic risk-free interest rate
 - r_f the foreign risk-free interest rate
- A foreign currency may be regarded as an investment asset that provides a yield equal to the foreign risk-free interest rate

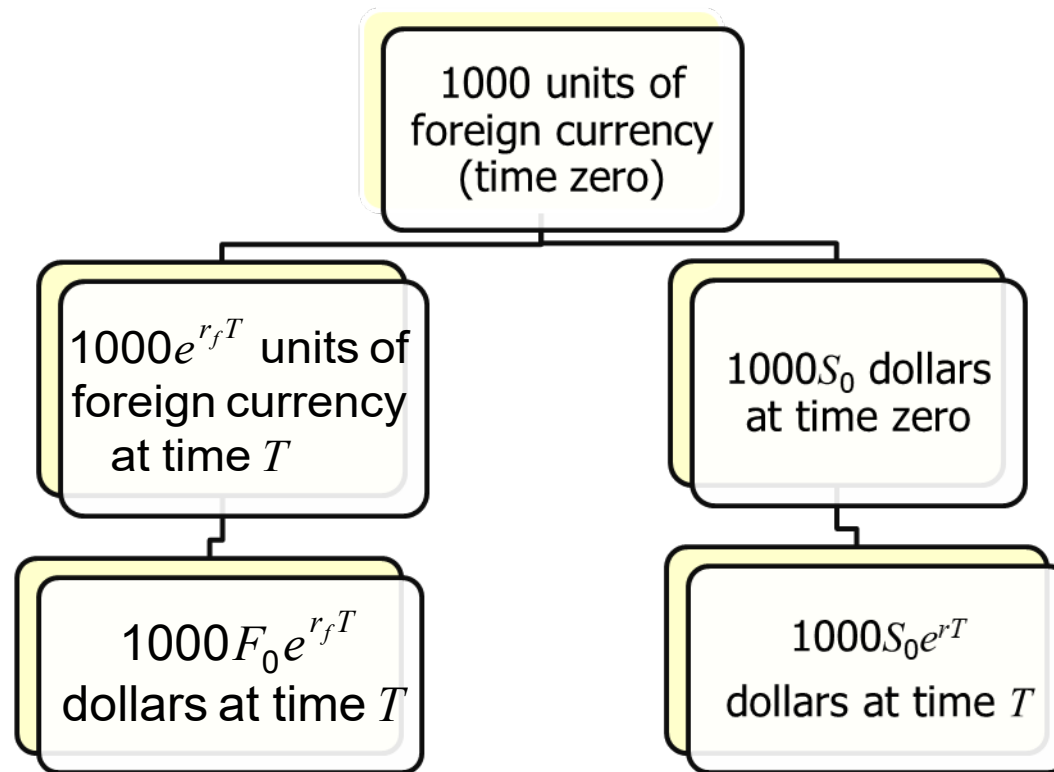
$$F_0 = S_0 e^{(r-r_f)T}, \quad F_t = S_t e^{(r-r_f)(T-t)}$$

Example. Suppose that the 2-year interest rates are 3% and 1% in Australia and US, respectively, and the spot exchange rate $S_0 = 0.75$ USD/AUD. Then the 2-year futures exchange rate

$$F_0 = S_0 e^{(r-r_f)T} = 0.75 e^{(0.01-0.03) \times 2} = 0.7206$$

Currency Futures

Two ways of converting currency



Appendix: Additional Reading

Derivatives Markets

Exchange Traded Derivatives
(e.g. futures, options)

Over-the-Counter Derivatives
(OTC)
(e.g. forwards, swaps)



Equities (S&P, SPI, Nasdaq)

Currencies (AUD/USD, Euro/USD)

Interest Rates (Treasury bonds)

New financials (credit, insurance)

Metals (gold, silver, copper)

Energy (crude oil, natural gas)

Softs (sugar, orange juice, corn)

Grains (corn, soybean, wheat)

Livestock (cattle, lean hog)

New commodities
(weather, electricity)

Forward

Example: On March 21, the treasurer of a corporation enters into a long forward contract to buy £1 million in six months at an exchange rate of 1.2230

- This obligates the corporation to pay \$1,223,000 for £1 million on September 21
- What are the possible outcomes?
 - If the exchange rate is x , the profit from the forward:
$$\$1m \times (x - 1.223)$$
 - But the cost is locked in
$$\$1m \times [x - (x - 1.223)] = \$1,223,000$$

Futures

- Delivery arrangements:
 - The exchange specifies where delivery occurs.
 - One gets to choose between a number of delivery locations.
- Delivery months:
 - The exchange specifies when trading in a particular month's contract may begin and the last day on which trading can occur during the delivery month.
- Price quotes:
 - The exchange defines how prices are quoted, which may not be the “actual” price (i.e., yield)
- Price limits, position limits, and circuit breaks
 - To prevent large price movement
 - To prevent large market speculation

Currency Futures

- Exchange rate:
 - The units of S_0 and F_0 are “units of the domestic currency per unit of the foreign currency.”
 - In the following, USD is treated as the domestic currency
 - For example, $S_0 = 0.16 \text{ USD/CNY}$, or, 1 CNY=0.16 USD
 - Note: This may differ from how exchange rates are quoted in the market.
- Currency futures:
 - The underlying asset for a currency futures contract is one unit of a foreign currency, e.g. 1CNY.

Long Hedge

- A long hedge is appropriate when the hedger knows he **expects to purchase** the underlying asset sometime in the future
- It involves a **long position in futures contracts**.
- **Example:** On 15 January a producer knows he **will require** 100,000 pounds of copper on 15 May.
 - NYMEX copper futures are for the delivery of 25,000 pounds.
 - The May contract is trading for \$3.2 per pound on 15 January.
 - The producer takes a **long position** in 4 contracts on 15 January to hedge his exposure
 - If S_T is the spot price of copper on 15 May and F_T is the May futures price on that date, then $F_T \approx S_T$.
 - On 15 May the producer pays $100,000 S_T - 4 \times 25,000 (F_T - 3.20) \approx \$320,000$:
 - So he has locked in a price of \$3.2 per pound.
 - This improves on the unhedged position if $F_T > \$3.2$, but it is worse if $F_T < \$3.2$.

